

CareBridge

Healthcare Referral & Settlement Platform

API Reference

Endpoint catalogue, authentication model, error handling, and real-time events for the CareBridge REST interface

DOCUMENT REFERENCE	CB-API-2026-004
VERSION	1.0
DATE OF ISSUE	2026-07-18
PREPARED FOR	CareBridge Health
PREPARED BY	CareBridge Engineering Team
CLASSIFICATION	Confidential

Table of Contents

1 Introduction	3
1.1 Scope	3
1.2 Base Address and Versioning	3
1.3 Conventions Used in This Document	3
2 Authentication Model	4
2.1 Credentials	4
2.2 Obtaining and Renewing a Session	4
2.3 Sign-In Preconditions	4
2.4 Rate Limiting	5
3 Response and Error Conventions	6
3.1 Response Shape	6
3.2 Status Codes	6
4 Authentication Endpoints	7
5 Referral Endpoints	8
5.1 Creating a Referral	8
6 Hospital Endpoints	10
6.1 Completing and Billing a Case	10
7 Laboratory Endpoints	12
7.1 Laboratory Management	12
7.2 Laboratory Referrals	12
7.3 Laboratory Administration	13
8 Settlement Endpoints	14
8.1 Hospital Settlements	14
8.2 Laboratory Settlements	14
9 Administration Endpoints	16
9.1 Users and Approvals	16
9.2 Party Registries and Commercial Terms	16
9.3 Platform Configuration and Operations	17
10 Profile, Notification, Upload, Export, and Payment Endpoints	19
10.1 Profile	19
10.2 Notifications	19
10.3 File Upload	19
10.4 Document Export	20
10.5 Payments	21
11 Real-Time Interface	22
11.1 Connecting	22
11.2 Rooms	22
11.3 Events	22
12 Data Conventions	24
12.1 Monetary Values	24
12.2 Enumerated Values	24
12.3 Identifiers and Formats	25

1. Introduction

1.1 Scope

This document catalogues the CareBridge HTTP interface. It is intended for engineers maintaining the platform or building an additional client against it. Every endpoint is listed with its method, path, the role required to call it, and its purpose.

1.2 Base Address and Versioning

All endpoints are served under a single version prefix. Paths in this document are written relative to that prefix — an entry shown as POST /referrals is called at https://<backend-host>/v1/referrals.

Base address	https://<backend-host>/v1
Content type	application/json (multipart/form-data for file upload)
Authentication	Authorization: Bearer <access token>
Health check	GET https://<backend-host>/ -> "CareBridge API is running"
Static files	https://<backend-host>/uploads/<filename>

1.3 Conventions Used in This Document

Column	Meaning
Method and path	The HTTP method and the path relative to the version prefix. A segment shown as :id is a path parameter.
Role	The minimum role required. See the note below on administrator access.
Purpose	What the endpoint does.

ADMINISTRATORS CAN REACH EVERY ENDPOINT

The authorisation guard permits the administrator role unconditionally, before the endpoint's own role list is consulted. The role shown against each endpoint below is therefore the minimum role, not an exclusive one. Every endpoint in this document is callable by an administrator. This is deliberate, to support platform oversight, but it means an administrator credential is an unrestricted credential.

2. Authentication Model

2.1 Credentials

Credential	Lifetime	Contents	Used for
Access token	1 hour	User identity, role, name, and creating account	Every authenticated request, sent in the Authorization header
Refresh token	30 days	User identity only	Obtaining a new pair of tokens when the access token expires

2.2 Obtaining and Renewing a Session

```
POST /v1/auth/login
{ "email": "doctor@example.com", "password": "..." }

200 OK
{
  "success": true,
  "accessToken": "<jwt>",
  "refreshToken": "<jwt>",
  "user": { "id": "...", "role": "consultant", "name": "..." }
}

--- when the access token expires ---

POST /v1/auth/refresh
{ "refreshToken": "<jwt>" }

200 OK -> a NEW access token AND a NEW refresh token
Store both. Renewal is rolling; the previous refresh
token should be discarded.
```

Renewal is refused if the account is no longer active. Suspending an account therefore ends its sessions at the next renewal, within one hour.

THERE IS NO CREDENTIAL REVOCATION LIST

A refresh token remains cryptographically valid for its full thirty-day lifetime. The only server-side control is the account status check performed at renewal. If a credential is believed to be compromised, suspend the account — that is the effective revocation mechanism.

2.3 Sign-In Preconditions

Condition	Result
Email not verified	Sign-in refused, with an indication that verification is outstanding so the client can offer to resend the code.

Condition	Result
Account pending approval	Sign-in refused. The account exists but has not been activated by an administrator.
Account suspended	Sign-in refused.
Account active and verified	Sign-in succeeds; a token pair is issued and a sign-in notification is raised.

2.4 Rate Limiting

Authentication endpoints are limited to thirty requests per minute per client address. The limit covers registration, sign-in, renewal, one-time-password issue and verification, and password reset. Exceeding it returns HTTP 429.

RATE LIMITING IS NOT APPLIED ELSEWHERE
No other router carries a limit, including the file upload endpoint and the administrative surface. Adding a global limit is recorded as a recommendation in the Security and Compliance document.

3. Response and Error Conventions

3.1 Response Shape

```
Success
{ "success": true, "data": { ... } }

Failure
{ "success": false, "message": "Human-readable explanation" }
```

3.2 Status Codes

Code	Meaning	Typical cause
200	Success	The request was processed.
201	Created	A new record was created.
400	Bad request	Validation failed — a missing field, a malformed identity or telephone number, or an invalid state transition.
401	Unauthenticated	The token is absent, malformed, or expired. The client should renew and retry once.
403	Forbidden	Authenticated, but the role is not permitted to call this endpoint.
404	Not found	The referenced record does not exist, or is not visible to this caller.
409	Conflict	A uniqueness constraint was violated — a duplicate email or professional registration number.
429	Too many requests	The authentication rate limit was exceeded.
500	Server error	An unhandled condition. The response body carries no internal detail.
502	Upstream failure	A third-party service failed — most commonly email delivery when issuing a verification code.

THERE IS NO CENTRAL ERROR HANDLER

Errors are handled within individual controllers rather than by a single application-level handler. Response shapes are consistent in practice but are not enforced structurally. Introducing a central handler is recorded in the Maintenance and Support Guide.

4. Authentication Endpoints

Base path `/auth`. These endpoints are public and are rate limited.

Method and path	Purpose
POST <code>/auth/register</code>	Register a consultant, hospital, or laboratory. Creates the login and the role profile, issues an email verification code, and notifies all administrators. Administrator registration is refused.
POST <code>/auth/login</code>	Authenticate and issue a token pair. Subject to the preconditions in section 2.3.
POST <code>/auth/refresh</code>	Exchange a refresh token for a new pair. Refused if the account is not active.
POST <code>/auth/verify-email-otp</code>	Verify an email address using the six-digit code.
POST <code>/auth/resend-email-otp</code>	Issue a fresh code. Returns 502 if delivery fails.
GET <code>/auth/verify-email</code>	Legacy link-based email verification, retained for compatibility.
POST <code>/auth/resend-verification</code>	Legacy link-based verification resend.
POST <code>/auth/forgot-password</code>	Issue a single-use, time-limited password reset link. Always returns success, so that registered addresses cannot be enumerated.
POST <code>/auth/reset-password</code>	Set a new password using a reset token, and notify the user.
GET <code>/auth/platform-settings</code>	Public branding — platform name, logo, colours, and favicon. Called before sign-in so the login page carries the operator's branding.

5. Referral Endpoints

Base path `/referrals`. All require authentication.

Method and path	Role	Purpose
GET <code>/referrals/suggestions</code>	consultant	Return hospitals ranked for a prospective referral by the matching engine.
POST <code>/referrals</code>	consultant	Create a referral. Allocates a referral code, sets the response deadline from urgency, and persists the ranked escalation queue.
GET <code>/referrals/mine</code>	consultant	The consultant's own referrals.
GET <code>/referrals/earnings</code>	consultant	Earnings summary and commission breakdown.
POST <code>/referrals/withdraw</code>	consultant	Request a withdrawal, subject to the minimum amount and available balance.
GET <code>/referrals/hospitals/:id/doctors</code>	consultant, hospital	Doctors at a hospital, for selecting a named recipient.
GET <code>/referrals/inbox</code>	hospital	Referrals awaiting this hospital's response.
GET <code>/referrals/hospital-all</code>	hospital	All referrals ever directed to this hospital.
PATCH <code>/referrals/:id/accept</code>	hospital	Accept a referral. A department assignment is required.
PATCH <code>/referrals/:id/reject</code>	hospital	Reject a referral. A reason is required and is shown to the consultant.
PATCH <code>/referrals/:id/status</code>	hospital	General status transition, underlying the two endpoints above.
GET <code>/referrals/:id</code>	consultant, hospital	Full referral detail including the clinical notes timeline.
PATCH <code>/referrals/:id</code>	consultant	Edit a referral. Permitted only while pending, accepted, or rejected.
POST <code>/referrals/:id/notes</code>	consultant, hospital	Append a clinical note, typed as nursing or consultant.

5.1 Creating a Referral


```
POST /v1/referrals
{
  "patientName": "...",
  "cnic": "35201-1234567-1",           // encrypted at rest
  "dateOfBirth": "1985-04-12",        // source of truth for age
  "gender": "male",                   // male | female | other
  "guardianRelation": "S/O",          // S/O | D/O | W/O
  "guardianName": "...",
  "phone": "03001234567",
  "area": "Gulshan-e-Iqbal",
  "urgency": "urgent",                // emergency | urgent | routine
  "symptomsText": "chest pain, breathlessness",
  "summaryNotes": "...",
  "attachments": ["https://..."],
  "targetHospitalId": "...",
  "department": "Cardiology",
  "targetDoctorId": "...",            // optional
  "rankedHospitalPreferences": [      // the escalation queue
    { "hospitalId": "...", "department": "Cardiology" }
  ]
}
```

201 Created – the response carries the allocated referral code and the response deadline derived from urgency.

6. Hospital Endpoints

Base path `/hospitals`. The whole router requires the hospital role.

Method and path	Purpose
GET <code>/hospitals/dashboard</code>	Headline figures for the hospital dashboard.
GET <code>/hospitals/analytics</code>	Referral volume over time, for the dashboard chart.
GET <code>/hospitals/financial-ledger</code>	Billed cases with amount billed, platform charge, and net position.
GET <code>/hospitals/referrals-pipeline</code>	Accepted referrals not yet admitted.
GET <code>/hospitals/beds</code>	Bed inventory by ward.
PATCH <code>/hospitals/beds</code>	Update ward bed counts.
PATCH <code>/hospitals/departments</code>	Update the departments the hospital operates. This determines referral eligibility.
GET <code>/hospitals/doctors</code>	The hospital's doctor register.
POST <code>/hospitals/doctors</code>	Add a doctor.
PATCH <code>/hospitals/doctors/:id</code>	Update a doctor, including the availability flag.
DELETE <code>/hospitals/doctors/:id</code>	Remove a doctor.
GET <code>/hospitals/admissions</code>	Current and past admissions.
POST <code>/hospitals/admissions</code>	Admit an accepted referral. Ward, room, bed, department, and treating doctor are all required.
PATCH <code>/hospitals/admissions/:id</code>	Amend admission details.
POST <code>/hospitals/admissions/:id/complete</code>	Bill and close a case. Computes the commercial split, closes the referral, and accrues the consultant commission. Idempotent.
GET <code>/hospitals/users</code>	The hospital's team member accounts.
POST <code>/hospitals/users</code>	Create a team member account.
DELETE <code>/hospitals/users/:id</code>	Remove a team member, subject to the creation-chain check.

6.1 Completing and Billing a Case

```
POST /v1/hospitals/admissions/:id/complete
{
  "services": [
    { "description": "Room charges (3 nights)", "amountPaisa": 4500000 },
    { "description": "Consultant visit", "amountPaisa": 800000 },
    { "description": "Investigations", "amountPaisa": 1200000 }
  ],
  "paymentMethod": "cash",
  "paymentReference": "RCPT-8841",
  "patientBillFileUrl": "https://..."
}
```

All amounts are WHOLE PAISA. 4500000 paisa = PKR 45,000.

Effects, in order:

1. bill total computed from the service lines
2. commercial split computed by the commission engine
3. admission marked billed; referral closed
4. consultant commission accrued, with a terms snapshot
5. STATUS_UPDATE emitted to the consultant and the hospital

Repeating this call on an already-billed admission returns without further financial effect.

7. Laboratory Endpoints

7.1 Laboratory Management

Base path `/labs`. The whole router requires the laboratory role.

Method and path	Purpose
GET <code>/labs/me</code>	The laboratory's own profile.
GET <code>/labs/dashboard</code>	Referral counts by status.
GET <code>/labs/tests</code>	The test catalogue.
POST <code>/labs/tests</code>	Add a test with price and turnaround time.
PATCH <code>/labs/tests/:testId</code>	Update a catalogue entry.
DELETE <code>/labs/tests/:testId</code>	Remove a catalogue entry.
GET <code>/labs/users</code>	Team member accounts.
POST <code>/labs/users</code>	Create a team member account.
DELETE <code>/labs/users/:id</code>	Remove a team member.

7.2 Laboratory Referrals

Base path `/lab-referrals`. Roles vary by endpoint.

Method and path	Role	Purpose
GET <code>/lab-referrals/suggestions</code>	consultant	Suggested laboratories for a prospective referral.
POST <code>/lab-referrals</code>	consultant	Create a laboratory referral with an ordered test list and an optional discount, capped for that consultant.
GET <code>/lab-referrals/mine</code>	consultant	The consultant's laboratory referrals.
PATCH <code>/lab-referrals/:id/re-refer</code>	consultant	Redirect a rejected referral to a different laboratory, preserving the patient details.
GET <code>/lab-referrals/inbox</code>	laboratory	Referrals awaiting a response.
GET <code>/lab-referrals/lab-all</code>	laboratory	All referrals directed to this laboratory.
PATCH <code>/lab-referrals/:id/accept</code>	laboratory	Accept a referral. An expected report date and time is required.
PATCH <code>/lab-referrals/:id/reject</code>	laboratory	Reject a referral with a reason.
POST <code>/lab-referrals/:id/reports</code>	laboratory	Upload report files. Notifies the consultant.

Method and path	Role	Purpose
PATCH /lab-referrals/:id/bill	laboratory	Price the ordered tests. The test list itself cannot be altered.
PATCH /lab-referrals/:id/finalize	laboratory	Close and bill the referral. Computes the split and accrues the consultant commission.
GET /lab-referrals/:id	consultant, laboratory	Full referral detail.

7.3 Laboratory Administration

Base path /admin/labs, requiring the administrator role. This router is mounted ahead of the general administrative router so that it takes precedence on the shared prefix.

Method and path	Purpose
GET /admin/labs	All laboratories.
GET /admin/labs/:id	One laboratory in detail.
PATCH /admin/labs/:id	Amend a laboratory profile.
PATCH /admin/labs/:id/status	Activate, suspend, or set pending.
GET /admin/labs/:id/consultant-overrides	Per-consultant commercial terms at this laboratory.
POST /admin/labs/:id/consultant-overrides	Set a per-consultant override.
GET /admin/labs/referrals	All laboratory referrals across the platform.
PATCH /admin/labs/referrals/:id	Correct a laboratory referral.
GET /admin/labs/payouts	Laboratory commission accruals.

8. Settlement Endpoints

8.1 Hospital Settlements

Base path `/settlements`. Roles vary by endpoint.

Method and path	Role	Purpose
GET <code>/settlements/pending-admissions</code>	hospital	Billed admissions not yet drawn into a settlement.
POST <code>/settlements</code>	hospital	Create a settlement over a stated period, with a summary document.
POST <code>/settlements/:id/upload-receipt</code>	hospital	Upload the payment receipt. Advances the settlement to pending verification.
GET <code>/settlements/hospital</code>	hospital	The hospital's settlement history.
GET <code>/settlements/admin</code>	admin	All settlements awaiting administrative action.
POST <code>/settlements/admin/:id/verify</code>	admin	Verify a receipt, or reject it with a reason that returns it to the hospital.
POST <code>/settlements/admin/:id/payout</code>	admin	Record a consultant payout with its receipt.
GET <code>/settlements/consultant</code>	consultant	Settlements carrying a payout to this consultant.
POST <code>/settlements/consultant/:id/verify</code>	consultant	Confirm receipt of a payout. Completes the settlement when all payouts are confirmed.

8.2 Laboratory Settlements

Base path `/lab-settlements`. The state machine is identical to the hospital cycle.

Method and path	Role	Purpose
GET <code>/lab-settlements/pending-referrals</code>	laboratory	Closed referrals not yet settled.
POST <code>/lab-settlements</code>	laboratory	Create a settlement over a stated period.
POST <code>/lab-settlements/:id/receipt</code>	laboratory	Upload the payment receipt.
GET <code>/lab-settlements/mine</code>	laboratory	The laboratory's settlement history.
GET <code>/lab-settlements/admin</code>	admin	Settlements awaiting administrative action.
POST <code>/lab-settlements/:id/verify</code>	admin	Verify or reject a receipt.
POST <code>/lab-settlements/:id/payout</code>	admin	Record a consultant payout.

Method and path	Role	Purpose
GET /lab-settlements/consultant	consultant	Settlements carrying a payout to this consultant.
GET /lab-settlements/consultant/earnings	consultant	Laboratory commission earnings summary.
POST /lab-settlements/:id/consultant-verify	consultant	Confirm receipt of a payout.

THE TWO SETTLEMENT ROUTERS HAVE DIFFERENT PATH SHAPES

Hospital administrative actions are namespaced under /settlements/admin/:id/..., while the laboratory equivalents are flat at /lab-settlements/:id/.... The two state machines are identical; only the paths differ. This is recorded as an inconsistency to be aligned in a future interface version, and is noted here so that it is not mistaken for a behavioural difference.

9. Administration Endpoints

Base path `/admin`. The whole router requires the administrator role.

9.1 Users and Approvals

Method and path	Purpose
GET <code>/admin/users</code>	All accounts.
GET <code>/admin/users/pending</code>	The registration approval queue.
PATCH <code>/admin/users/:id</code>	Change account status — activate, suspend, or set pending.
POST <code>/admin/users/:id/change-password</code>	Set a new password for a user.
DELETE <code>/admin/users/:id</code>	Delete an account permanently.
GET <code>/admin/admins</code>	Administrator accounts.
POST <code>/admin/admins</code>	Create an administrator account.
DELETE <code>/admin/admins/:id</code>	Remove an administrator account.

9.2 Party Registries and Commercial Terms

Method and path	Purpose
GET <code>/admin/consultants/:id/profile</code>	Full consultant profile.
GET <code>/admin/consultants/:id/patients</code>	Patients referred by this consultant.
PATCH <code>/admin/consultants/:id</code>	Amend a consultant profile.
POST <code>/admin/consultants/:id/commission</code>	Set this consultant's commission model, type, and rates.
GET <code>/admin/hospitals/:id/patients</code>	Patients treated at this hospital.
PATCH <code>/admin/hospitals/:id</code>	Amend a hospital profile.
POST <code>/admin/hospitals/:id/deduction</code>	Set the hospital's platform charge type and value.
GET <code>/admin/hospitals/:id/consultant-overrides</code>	Per-consultant terms at this hospital.
POST <code>/admin/hospitals/:id/consultant-overrides</code>	Set a per-consultant override.
GET <code>/admin/hospitals/:id/doctors</code>	The hospital's doctor register.

Method and path	Purpose
POST /admin/hospitals/:id/doctors	Add a doctor on the hospital's behalf.
PATCH /admin/hospitals/:id/doctors/:doctorId	Amend a doctor record.
DELETE /admin/hospitals/:id/doctors/:doctorId	Remove a doctor record.

9.3 Platform Configuration and Operations

Method and path	Purpose
GET /admin/analytics	Platform-wide analytics.
GET /admin/scoring	The current matching weights.
PUT /admin/scoring	Set the matching weights. Rejected unless they total exactly 100.
GET /admin/settings	Platform financial defaults and branding.
PUT /admin/settings	Update platform settings.
GET /admin/departments	The department and keyword catalogue.
POST /admin/departments	Add a department with its symptom keywords.
PATCH /admin/departments/:id	Amend a department entry.
DELETE /admin/departments/:id	Remove a department entry.
GET /admin/referrals	All referrals across the platform.
PATCH /admin/referrals/:id	Correct any referral, including admission details.
PATCH /admin/referrals/:id/override	Administrative override of referral routing.
DELETE /admin/referrals/:id	Delete a referral permanently.
GET /admin/beds	Bed inventory across all hospitals.
PATCH /admin/beds/:hospitalId	Correct a hospital's bed counts.
GET /admin/admissions	All admissions.
GET /admin/payouts	Consultant commission accruals.
PATCH /admin/payouts/:payoutId	Mark a payout as paid.
GET /admin/audit-logs	The audit log.
GET /admin/audit-logs/export	Export the audit log.
GET /admin/whatsapp/status	Messaging integration status.

Method and path	Purpose
GET /admin/whatsapp/users	Recipients reachable by message.
POST /admin/whatsapp/send	Send a broadcast message.

THE MESSAGING CONSOLE IS BUILT BUT NOT ROUTED IN THE CLIENT

The three messaging endpoints are live and callable. The administrative screen that uses them exists in the client codebase but is not linked into the application's navigation, so the feature is currently reachable only through the API.

10. Profile, Notification, Upload, Export, and Payment Endpoints

10.1 Profile

Base path /profile. Available to any authenticated role.

Method and path	Purpose
GET /profile/me	The caller's own profile, including role-specific detail and branding.
PUT /profile/me	Update the caller's own profile.
POST /profile/documents	Attach a verification document. Documents already verified by an administrator cannot be replaced.
POST /profile/favorites	Add or remove a favoured hospital, feeding the preference score.
POST /profile/change-password	Change the caller's own password.

10.2 Notifications

Base path /notifications. Available to any authenticated role.

Method and path	Purpose
GET /notifications	Recent notifications for the caller.
GET /notifications/unread-count	The unread counter shown in the interface.
PATCH /notifications/:id/read	Mark one notification read.
PATCH /notifications/read-all	Mark all notifications read.

10.3 File Upload

```
POST /v1/upload
Content-Type: multipart/form-data
field name: file

Accepted types  jpeg, jpg, png, pdf
Maximum size    5 MB
Validation      both file extension and declared content type

200 OK
{ "success": true, "url": "https://res.cloudinary.com/..." }

Files are stored in Cloudinary when configured, otherwise on
the local filesystem and served from /uploads.
```

THIS ENDPOINT DOES NOT REQUIRE AUTHENTICATION

It is the only unauthenticated write endpoint besides the payment gateway callback. Uploads are constrained by type and size, and an uploaded file only becomes meaningful once an authenticated user attaches it to a record — but the endpoint itself is open. Placing it behind authentication is the highest-priority security recommendation in this handover package.

10.4 Document Export

Base path /exports. All endpoints return a generated PDF document.

Method and path	Role	Produces
GET /exports/consultant/referrals	consultant	The consultant's referral roster.
GET /exports/consultant/referrals/:id	consultant	One referral record, with attachments merged in.
GET /exports/consultant/lab-referrals	consultant	The consultant's laboratory referral roster.
GET /exports/consultant/lab-referrals/:id	consultant	One laboratory referral record.
GET /exports/hospital/records	hospital	The hospital's referral records.
GET /exports/hospital/referrals/:id	hospital	One referral record.
GET /exports/lab/records	laboratory	The laboratory's referral records.
GET /exports/lab/referrals/:id	laboratory	One laboratory referral record.
GET /exports/admin/referrals/:id	admin	Any referral record.
GET /exports/admin/consultants/:id	admin	A consultant profile file.
GET /exports/admin/hospitals/:id	admin	A hospital profile file.
GET /exports/admin/lab-referrals/:id	admin	Any laboratory referral record.
GET /exports/admin/laboratories/:id	admin	A laboratory profile file.

10.5 Payments

Method and path	Access	Purpose
GET /payments/initiate-jazzcash/:admissionId	Any authenticated user	Build a signed payment request for an admission. No role check is applied.
POST /payments/jazzcash-callback	Public	Receive the gateway result. Secured by cryptographic signature verification rather than by authentication.

The signing scheme sorts the request parameters, concatenates the integrity salt with the non-empty values, and produces a keyed hash that the gateway verifies. Callbacks are verified by recomputing the same digest, so an unsigned or tampered callback is rejected. Per-hospital merchant credentials override the platform defaults where present.

11. Real-Time Interface

11.1 Connecting

```
const socket = io(SOCKET_URL, { transports: ["websocket"] });

socket.on("connect", () => {
  socket.emit("join", { token }); // user:<id> + role:<role>
  socket.emit("join_hospital", { token }); // facility rooms
  socket.emit("join_consultant", { token });
  socket.emit("join_laboratory", { token });
});
```

Each join verifies the token, confirms the role matches, resolves the facility, and joins the room. A failed join is logged and ignored; it does not close the connection.

11.2 Rooms

Room	Members	Receives
user:<id>	One user	Personal notifications
role:<role>	All users of a role	Role-wide broadcasts
hospital:<id>	A hospital owner and its team	Inbox, escalation, bed, and clinical note events
consultant:<id>	One consultant	Referral status, escalation, and settlement events
lab:<id>	A laboratory owner and its team	Laboratory referral and report events

11.3 Events

Event	Raised when	Sent to
notification	Any notification is raised	The recipient's personal room
NEW_REFERRAL	A referral is created or escalated into a facility	The receiving hospital room
STATUS_UPDATE	A referral changes status, including on billing	The consultant and hospital rooms
REFERRAL_ESCALATE D	A referral passes its deadline and is reassigned	The previous hospital and the consultant
BED_UPDATE	Bed counts change	The hospital room
NEW_CLINICAL_NOTE	A clinical note is added	The consultant and hospital rooms

Event	Raised when	Sent to
NEW_LAB_REFERRAL	A laboratory referral is created	The receiving laboratory room
LAB_STATUS_UPDATE	A laboratory referral changes status	The consultant and laboratory rooms
LAB_REPORT_UPLOAD DED	A report file is uploaded	The consultant room

The reference client subscribes to all nine events and collapses them into a single cache invalidation after a short delay, so a burst of related events produces one refresh. A client implementing its own handling should adopt a similar approach.

12. Data Conventions

12.1 Monetary Values

EVERY MONETARY VALUE IS A WHOLE NUMBER OF PAISA

One rupee is one hundred paisa. A field carrying PKR 45,000 holds the integer 4500000. Field names carrying money end in Paisa. Never send a decimal, a formatted string, or a rupee value to a paisa field — the conversion boundary is the client, and the API accepts integers only.

12.2 Enumerated Values

Field	Permitted values
User role	consultant, hospital, laboratory, admin
Account status	pending, active, suspended
Referral urgency	emergency, urgent, routine
Referral status	pending, accepted, rejected, admitted, closed
Laboratory referral status	pending, accepted, reported, closed, rejected
Admission status	active, discharged, billed
Settlement status	pending_payment, pending_admin_verification, paid_pending_consultant_payout, paid_pending_consultant_verification, completed
Payment method	pending, cash, jazzcash, easypaisa, bank_transfer, manual
Ward type	General, Private, ICU, NICU, PICU, HDU, Burns, Maternity, Psychiatric, Cardiac
Gender	male, female, other
Guardian relation	S/O, D/O, W/O
Clinical note type	nursing, consultant
Commission model	legacy, additive
Commission and charge type	percentage, fixed
Payout status	accrued, paid, pending_withdrawal

12.3 Identifiers and Formats

Item	Format
Record identifiers	24-character hexadecimal database identifiers.
Referral code	CB-YYYY-NNNN, allocated atomically and sequentially within the year.
Laboratory referral code	LAB-YYYY-NNNN, allocated from a separate sequence.
Dates and times	ISO 8601. Times in message templates are rendered in Pakistan Standard Time.
National identity number	Five digits, hyphen, seven digits, hyphen, one digit. Encrypted before storage on referral records.
Telephone number	Validated against Pakistani mobile and landline formats.
Geographic coordinates	A two-element array ordered longitude then latitude, following the geospatial standard.

COORDINATE ORDER

Geographic points are stored longitude first, then latitude — the order required by the geospatial standard, and the reverse of how coordinates are usually spoken. Reversing them places a hospital in the wrong hemisphere and silently corrupts every distance calculation for that record.